

MOD 302 Course Outline

MOD 302 — Input-Output and Structural Analysis

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	27 (indicative band 24–30)
Shape	4 parts · 8 sessions · 5 laboratories · 4 assignments · 1 examination · 1 final project
Delivery	Instructor-led online, blended, or intensive in-person
Prerequisites	MOD 301, or a working knowledge of social accounting matrices. Matrix algebra to the level of inversion. No prior input-output experience is assumed.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

An input-output table is a social accounting matrix read as a technology. This course takes the balanced SAM of MOD 301 and asks what it already implies about the structure of production: which sectors pull output through the economy, which ones everything else depends on, where a shock propagates and where it stops. It then moves to a 133-sector multi-regional table and does the same work at a resolution where the answers become specific — including the answers that turn out to be artefacts of an almost empty sector.

The course runs on two data layers and refuses to mix them: the balanced Utopia SAM of MOD 301 read as a Leontief system, and a 133-sector multi-regional table covering the focus economy and seven peers. Both of the findings the course is built around are reproduced by learners rather than asserted — two of the three key sectors in the reference economy are the two smallest sectors in it, and nine sectors of the multi-regional table share one backward linkage to four significant figures. Both are prevented by a three-line screen, and both are in the 51 registered findings the appendix carries.

The rule the course is built on

A number that has not been screened is not a result. Input-output arithmetic will return a number for any table you give it, including a table whose sector is empty. A multiplier computed on a near-zero base is arithmetically correct and economically meaningless, and nothing in the formula says so. The screen is not optional and it is not an afterthought: it is the step that decides whether there is a result.

Who it is for

Analysts in planning ministries, statistics offices, central banks and research institutes who have a balanced SAM or a published input-output table and need to say something defensible about economic structure.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	From matrix to technology	01–02	01	5.5
P2	What the inverse implies	03–04	02–03	7.0
P3	Other readings of the same table	05–06	04	6.0
P4	Production as a network, at high resolution	07–08	05	5.5
	Assignment and assessment time			3.0
	Total	8	5	27.0

The 8 sessions and 5 laboratories, in teaching order, and the subject matter of each:

P1 From matrix to technology — Sessions 01–02 · 5.5 h · Laboratory 01

- **01 From SAM to Input-Output Table** — The intermediate block, isolated; Dividing by output: the technical coefficient; What fixed coefficients assume, stated precisely; Industry and commodity technology, again; Domestic versus total requirements; The identity $x = Ax + f$, and where it comes from
- **02 The Leontief System** — Solving $x = Ax + f$; The power series, and why the inverse is a sum of rounds; Existence: the Hawkins-Simon conditions; Reading a column of L , and reading a row; The diagonal, and why it exceeds one; The identity check as a diagnostic
- *Lab 01 Build A and L from the SAM* — Can I get from a balanced SAM to a Leontief inverse, and check that I did it right?

P2 What the inverse implies — Sessions 03–04 · 7.0 h · Laboratory 02–03

- **03 Multipliers** — The output multiplier as a column sum; Value-added and income multipliers; Import leakage, and why it belongs beside the multiplier; Type I and Type II: closing the model on households; Employment multipliers and the data they need; What a multiplier is not: a recommendation
- **04 Linkages and Key Sectors** — Backward linkage from the Leontief inverse; Forward linkage from the Ghosh inverse; Normalisation, and what a value of one means; The four quadrants and their policy readings; The supply-driven quadrant: vulnerability, not opportunity; Rasmussen, Hirschman, and what the literature actually claimed
- *Lab 02 Multipliers and What They Measure* — What does each multiplier column actually answer, and which question is being asked?
- *Lab 03 Key Sectors and the Four Quadrants* — Which sectors are key, which are vulnerabilities, and what would I actually recommend?

P3 Other readings of the same table — Sessions 05–06 · 6.0 h · Laboratory 04

- **05 The Ghosh Model** — The allocation matrix B ; The Ghosh inverse $G = (I - B)^{-1}$; The price interpretation, and the quantity interpretation that does not survive; Dietzenbacher's reading; When a forward linkage is informative; Forward linkage exactly one: the empty-row signature

- **06 Structural Decomposition** — Two tables, two periods, one question; The additive decomposition and its non-uniqueness; The polar decompositions and the average of them; The interaction term, and reporting it separately; What SDA needs that a single table cannot give; Deflation: the step that decides the answer
- *Lab 04 Multipliers That Are Artefacts* — This sector has the highest multiplier in the economy. Is that a finding?

P4 Production as a network, at high resolution — Sessions 07–08 · 5.5 h · Laboratory 05

- **07 Production as a Network** — An input-output table is a weighted directed graph; Out-strength and in-strength: linkages by another name; Eigenvector centrality: importance by association; Betweenness: the sectors shocks travel through; What centrality adds beyond a linkage measure; Reading the Utopia network
- **08 Trade in Value Added, and Where This Goes Next** — Gross exports are not value added; The DVA and FVA decomposition; Backward GVC participation; Concentration: Herfindahl and the top-five share; Comparing eight countries without over-reading; Where MOD 308 takes the multi-regional table
- *Lab 05 Networks and Cross-Country Structure* — What does centrality add, and how far can I push a comparison across countries?

What a graduate can do

1. Derive the technical coefficient matrix and the Leontief inverse from a balanced SAM, and state the assumption each step makes.
2. Compute output, value-added, income and employment multipliers, and say precisely what each one does and does not measure.
3. Compute backward and forward linkages, classify sectors into the four quadrants, and explain why the supply-driven quadrant is the one policy discussion usually misses.
4. Identify a multiplier that is an artefact of a near-empty sector, and prove it is one rather than asserting it.
5. Apply the Ghosh supply-side model, state its interpretive difficulty, and say when it may and may not be used.
6. Decompose a change in output into technology, final demand and trade components, and report the interaction term separately.
7. Read a production system as a network: compute strength, eigenvector and betweenness centrality, and say what each adds beyond a linkage measure.
8. Decompose exports into domestic and foreign value added, compute backward GVC participation, and compare a country against its peers without over-reading the differences.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 80 items, used for formative practice and drawn on by the theory paper, which carries the marks

Component	Weight	What it is
Laboratory exercises	35 %	5 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The checked Leontief system; Four multipliers, four questions; The screened linkage table; The structural note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete structural analysis of an economy of the learner's choosing, on either data layer, delivered as a nine-section structural note with the code and the screened tables

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted microdata and no household record appears in this course. Both layers are published or licensed aggregate tables, and what ships is the derived coefficient and inverse matrices with the sector classification each was computed on. The two layers are never combined in a figure: different classifications, different years and different geographies, and mixing them is the most likely way this material is misused.

Materials and set-up

A 167-page Technical Manual, 9 lecture decks of 179 slides, 5 laboratory decks of 69 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 80-item quiz bank, 5 assessment files, 32 datasets, 4 code exercises, 4 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

The laboratories run in Julia against the shared course code and need no model instantiated: the input tables ship with the course. Learners who also want to rebuild the SAM layer from source need CDCGE at D:/EDIP/models/CDCGE and must start Julia with `-pkgimages=no`, for the reason school gap M-02 gives.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

MOD 302 Input-Output and Structural Analysis · version 1.0 · AICTEM EDIS Academy, School 03 Economic Modelling.